Multimodal AI Framework for Intelligent Caregiver Support and Patient Safety in Care Facilities

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***Abstract*—Hospital and care-facility caregivers face occupational stress, patient falls, heavy documentation, and communication barriers with deaf or non-verbal patients, yet each is usually addressed by a separate tool. This paper asks how well four machine learning modules, built for one web-based caregiver platform, perform when each is evaluated on held-out people or data. PULSE predicts stress risk from wearable, workload, and sleep data recorded before each survey, and tests whether sensor-measured proximity to colleagues adds value. SENTRY detects falls from webcam video by combining a Spatial-Temporal Graph Convolutional Network with biomechanical features. SCRIBE turns free spoken narration into structured Activities of Daily Living records with safety alerts, and GLOSS gives sign-language learners landmark-level feedback based on Dynamic Time Warping. On 27 held-out caregivers, PULSE reached a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.7211 (95% confidence interval 0.6644 to 0.7699), and 0.6991 in repeated participant-grouped cross-validation. Most of its signal came from the previous stress report; proximity added nothing detectable. SENTRY achieved 97.12% accuracy with a 1.28% false-alarm rate on unseen subjects; on a different dataset its ROC-AUC fell to 0.410, and webcam fine-tuning on one subject recovered a macro-F1 of 0.8851 on a second subject. SCRIBE reached a 15.57% Word Error Rate and caught 82.00% of alerts in a 200-case safety set. GLOSS recognized signs with a macro-averaged F1-score of 0.75, and 91.5% of the feedback rephrased by its large language model kept the rule-based content intact. Results are reported per module, including negative findings and limitations.**

***Index Terms*—Multimodal Deep Learning, Caregiver Support Systems, Domain Adaptation, Fall Risk Detection, Sign Language Recognition, Stress-Risk Prediction, Speech-to-Text Documentation**

I. Introduction

Caregivers in hospitals, eldercare, and long-term disability care work under several pressures at once, and four of them are the focus of this paper. Stress builds up from workload, working relationships, and physiological strain, but these are rarely modelled together. Falls can go unnoticed when monitoring depends on staff being present. Activities of Daily Living (ADL) records are still largely written by hand. Finally, few caregivers are trained in sign language, which makes communication with deaf or non-verbal patients difficult.

These pressures have measurable costs. About one-third of registered nurses in the United States (US) experience burnout, and the turnover it causes has been estimated to cost a hospital $16,736 a year for each nurse it employs [1]. More than 14 million US adults aged 65 and over fall each year. In 2023, falls caused 41,400 deaths in this age group, and the medical costs of non-fatal falls are about $80 billion a year [2]. Nurses spend up to 41% of their time on electronic health records and documentation [3]. In a survey of 238 health professionals in Riyadh, Saudi Arabia, only 6.3% had formal sign-language training, although 69.3% considered it very important for patient care [4].

Existing tools usually address one of these problems, and each line of work has its own gap. Stress models are often built on features measured on the same day as the stress report, and rarely test whether measured contact with colleagues adds anything [5]. Skeleton-based fall detectors are mostly trained and tested on data from the same laboratory, so their behaviour with a new camera or room is seldom reported [6]. Documentation tools for nurses often stop at transcription [7]. Sign-language tutors can compare a learner's sign with a reference [8], but showing which part of the sign went wrong has received less attention. To our knowledge, no system we reviewed brings stress, falls, documentation, and sign-language communication together in a single platform for caregivers.

This paper presents four modules developed for one web-based caregiver platform: PULSE (stress risk), SENTRY (fall

risk), SCRIBE (documentation), and GLOSS (sign-language communication). They share a backend and web interface, but each is trained and evaluated separately, because the datasets, participants, and deployment conditions differ. An end-to-end evaluation of the combined platform is outside the scope of this paper. The question addressed here is how well each module performs on people or data it did not see during training.

PULSE contributes a two-stage stress-risk pipeline built only from data available before each survey, in which a learned heart-rate baseline replaces a flat personal mean. Its proximity-centrality ablation spans three model families, and results are reported with caregiver-level confidence intervals and repeated grouped cross-validation. SENTRY is an edge-deployable fall detector that fuses a skeleton stream with biomechanical features. Its drop under dataset shift is measured and partly recovered through site-specific fine-tuning. SCRIBE turns free speech into structured ADL records, selecting models by alert recall rather than overall accuracy and testing them on an independent safety set. GLOSS extends sign recognition with speed-tolerant, landmark-level feedback, and checks whether the feedback rephrased by its large language model (LLM) stays faithful to the rule-based output. Related work is covered in Section II, the four modules in Section III, their results in Section IV, and our conclusions in Section V.

II. Literature Review

A. Theoretical and Technical Foundations

1) Stress Detection in Healthcare Workers: Much of the field evidence on continuous stress sensing in hospital staff comes from the Tracking Individual Performance with Sensors (TILES) programme. In TILES-2018, 212 hospital workers wore wearable sensors for about ten weeks, carried a Bluetooth badge detected by receivers placed around the hospital, and answered repeated Ecological Momentary Assessment (EMA) surveys [9]. The badge data make it possible to estimate who was near whom, which few stress datasets allow. Results on these cohorts also show how fragile stress prediction can be. Using heart rate and step count alone, one study reached a ROC-AUC of 0.84 on TILES-2019 but only 0.65 on a second real-world dataset [5], so a result from a single cohort should be read with caution. Two questions matter for an early-warning tool and have received limited attention. The first is whether stress can be predicted from data recorded before the report, which is all a deployed system would have. The second is whether measured contact with colleagues adds information once physiology and workload are known. PULSE is built around both questions.

2) Fall Detection and Domain Shift: Fall detection has moved from wearable and depth sensing toward ordinary cameras. An early embedded system used a worn accelerometer to flag a possible fall and a Kinect depth camera to confirm it, and its authors released the UR Fall Detection dataset [10]. The design works in controlled rooms but needs a depth sensor and a device on the patient. Skeleton-based methods avoid both by describing a person as a set of joints moving over time. Spatial-temporal graph convolutional networks (ST-GCN) treat the skeleton as a graph and learn motion patterns directly from it; they were introduced for general action recognition and evaluated on large datasets such as NTU RGB+D [11], [12]. Detectors built on such models are, however, usually trained and tested within one dataset recorded with one type of sensor. Studies of dataset bias show that models evaluated this way can learn the peculiarities of the collection and lose accuracy on data gathered elsewhere [6]. The risk is sharper for falls, because skeletons captured by laboratory depth sensors differ from skeletons estimated from webcam video by pose estimators such as BlazePose [13]. How much a skeleton-based detector degrades across datasets, and how much of that loss site-specific data can recover, is seldom reported. SENTRY measures both.

3) Speech-Based Documentation: Speech recognition for caregiving must cope with accents, mixed languages, and clinical vocabulary. General-purpose recognizers such as Whisper, trained with large-scale weak supervision [14], still perform unevenly on non-native speech: the Svarah benchmark, with 9.6 hours of English from 117 speakers across India, showed clear room for improvement in six open-source and two commercial systems [15]. Fine-tuning Whisper on Indian-accented English has been studied as a remedy [16], and Low-Rank Adaptation (LoRA) lets this tuning train a small set of added weights instead of the whole model [17]. In nursing, a Mandarin–English code-switching recognizer trained on about 100 hours of recordings reached a word error rate of 22.20%, rising to 31.06% on a clinical case [7]; that work ends at transcription. A more recent system passes transcripts from an adapted Whisper model to a large language model that writes structured nursing notes, which nurses check before upload [18]. SCRIBE shares the goal of structured records but differs in three respects: it targets Indian-accented English; it uses smaller models (Whisper-small and T5-small), which reduces model size and computational requirements compared with the larger models in that study; and it explicitly evaluates safety-alert detection alongside structured ADL extraction, rather than documentation quality alone.

4) Sign-Language Recognition and Tutoring: Isolated sign recognition has shifted from raw video to body and hand keypoints. Word-level datasets such as WLASL [19] and the community-sourced ASL Citizen [20] support this line of work, and compact pose-based models such as the SPOTER transformer classify signs from keypoint sequences alone [21]. Temporal convolutional networks offer a causal alternative that suits streaming input [22]. These recognizers return a label: they say which sign was made, not how well it was made. Tutoring systems add a comparison with a reference performance. A smartphone tutor for Chinese Sign Language, for example, evaluates a learner's sign against a 1,000-word vocabulary in about 700 ms per sample [8]. Its published description covers comparing the learner's sign with the reference and retrieving likely signs, but does not describe feedback that locates where a sign went wrong. GLOSS targets this gap between recognition and useful correction by locating execution errors at landmark level.

B. Comparison with Existing Systems

- **Wearable stress classifier (TILES-2019)** [5] uses heart rate and step count, reaching ROC-AUC 0.84 on TILES-2019 and 0.65 on a second dataset. *Gap:* proximity data are not used, and prediction from pre-report data alone is not examined.
- **Depth-and-accelerometer fall detector** [10] confirms accelerometer alarms with Kinect depth data on embedded hardware. *Gap:* it needs a depth sensor and a worn device, which RGB-only care rooms do not have.
- **ST-GCN skeleton models** [11] learn motion patterns from joint graphs. *Gap:* they are typically evaluated within one dataset, so cross-dataset loss for falls is not reported.
- **Indian-accent ASR benchmark (Svarah)** [15] evaluates eight recognizers on Indian-accented English. *Gap:* it measures transcription only, with no caregiving content or downstream structuring.
- **Code-switching nursing ASR** [7] transcribes Mandarin–English nursing records. *Gap:* output is plain text; no fields are extracted and no alerts are raised.
- **ASR with LLM nursing notes** [18] produces structured notes that nurses verify. *Gap:* it uses larger models on Mandarin–English speech, and its reported evaluation covers note accuracy and hallucination rather than safety-alert detection.
- **Pose-based sign recognizers (SPOTER)** [21] classify isolated signs from keypoints. *Gap:* they return a label, not an assessment of how well the sign was performed.
- **Smartphone sign tutor** [8] compares a learner's sign with a 1,000-word reference set in about 700 ms. *Gap:* its published description does not include feedback that localizes errors within the sign.

III. Methodology

A. PULSE: Personalized Stress-Risk Detection and Relational Network Analysis

PULSE tests whether caregiver stress can be predicted from data available before a stress report, and whether sensor-measured proximity to colleagues adds to physiological and workload information.

Data and target: TILES-2018 [9] provides 14,795 Ecological Momentary Assessment (EMA) prompts from 212 caregivers, of which 11,350 carry a stress rating (1 to 5); ratings of 3 or higher were labelled elevated. Before any feature was built, caregivers were shuffled once with a fixed seed into training (140), validation (30), and test (27) partitions, and 15 more were set aside for the application demonstration. No caregiver appears in more than one partition.

Features: Each prompt is described only with data available when it was sent: heart-rate mean, standard deviation, minimum, and maximum from earlier that day; 3-day and 7-day deviations from a personal baseline built on completed days; the previous day's steps, activity-zone minutes, and resting heart rate; the preceding night's sleep efficiency; and the previous stress rating (lag-1 stress). Steps deviation is the one exception, since its reference is the caregiver's whole-period mean, so a prior-days version was tested separately. The table holds 8,692 prompts (5,989 training, 1,296 validation, 1,407 test).

Two-stage model: Stage 1, a LightGBM regressor (100 trees, depth 3), predicts heart rate from the 3-day and 7-day trend and the day of the week. It has no participant identifier and is fitted on training caregivers only; its residual replaces the flat personal deviation. Stage 2 is a LightGBM classifier on the 16 features (depth 3, learning rate 0.05, L1 and L2 penalties 0.5, subsampling 0.8), weighted by the training class ratio (23.98% elevated) and stopped early on validation ROC-AUC. Missing values were median-imputed from training data. SHapley Additive exPlanations (SHAP) [23] were computed on the test predictions.

Evaluation: ROC-AUC is the primary metric; accuracy (0.5 threshold) is reported against the majority-class rate. Because prompts from one caregiver are not independent, 95% confidence intervals (CIs) come from 2,000 bootstrap resamples of test caregivers. The whole pipeline was also refitted in five repeats of 5-fold participant-grouped cross-validation on the 170 training and validation caregivers, with 15% of each training fold used for early stopping, and compared with lag-1 stress used alone.

Proximity ablation: In each 5-minute window, caregivers were placed at the badge receiver with the strongest signal, and those sharing a receiver were treated as co-located, giving daily degree (distinct colleagues) and weighted (co-location events) centrality. Centrality exists for only 2,989 of 10,038 daily records, so this earlier analysis used that subset (2,018/500/471) with same-day features. Random Forest (class-balanced), Extreme Gradient Boosting (XGBoost), and LightGBM were trained with 14 physiological and workload features (Set A) and with centrality added (Set B). A Welch t-test compared stress on high- and low-centrality days, and point-biserial correlations related elevated stress to heart-rate measures and sleep efficiency.

B. SENTRY: Skeletal-Movement-Based Fall-Risk Detection

SENTRY detects falls from an ordinary red-green-blue (RGB) webcam, without wearables or depth sensors, and runs on a CPU.

Data: Training draws on four action classes from NTU RGB+D [12]. Falling (A043) is the positive class, and sitting down (A008), standing up (A009), and jumping up (A027) serve as hard negatives. These are combined with the UR Fall Detection dataset [10], giving 578 labelled clips, of which 431 are normal and 147 are falls. NTU RGB+D supplies Kinect v2 skeletons, which we map to 14 joints, whereas UR Fall Detection is processed with MediaPipe Pose [13] from its RGB frames to resemble the deployment input; this difference in skeleton pipelines is likely a major source of the cross-dataset gap. For domain adaptation, we recorded 93 webcam clips: 35 from one subject for fine-tuning and 58 from a second subject for testing only. The decision threshold was calibrated on a

validation subset of the 35 fine-tuning clips, so the 58 test clips played no part in training or calibration.

Evaluation: Three protocols are used. In the main protocol, clips are split by subject, so nobody in the test set appears in training. A random 80/10/10 split at clip level gives an upper bound, since the same people can then appear on both sides. The third protocol trains on NTU RGB+D and tests on UR Fall Detection. We report accuracy, sensitivity, false-alarm rate, and ROC-AUC for the main protocol, accuracy and false-alarm rate for the random split, ROC-AUC for the cross-dataset test, and macro-F1 in addition for the webcam test.

Model: Each skeleton is reduced to 14 joints (from the 33 MediaPipe landmarks where applicable), low-visibility points are removed, and the skeleton is centered on the hip midpoint. Sequences are cut into 90-frame windows, or 3 s at 30 frames per second, and the features are scaled. Every window then passes through two branches. The first is an ST-GCN [11] with three graph temporal convolutional blocks of 32, 64, and 128 channels and a temporal kernel of 9, which outputs a 128-dimensional embedding. The second computes 18 handcrafted biomechanical features. One of them, F_{13}, combines torso sway and movement frequency as $0.60 \times \sigma(\text{torso angle}) + 0.40 \times$ oscillation rate, where σ is the standard deviation of the torso inclination angle within the window. Joining the 128 learned and 18 handcrafted values gives 146 inputs to a two-layer multilayer perceptron (MLP), which makes the final decision. Every network uses AdamW (learning rate 3×10^{-4}, weight decay 1×10^{-4}), batches of 32, dropout of 0.3, seed 42, up to 60 epochs, and a patience of 10 for early stopping. For webcam adaptation, only the last two fusion layers are updated, at a learning rate of 1×10^{-5}, and the calibrated decision threshold is 0.32. The ablation uses two baselines: a class-balanced Random Forest with 300 trees on the same 18 biomechanical features, and the ST-GCN branch on its own.

Live operation: At run time, a zone-aware context engine assigns the person one of five posture states and adds a score modifier when the person is inside a zone polygon drawn by a supervisor; ray casting decides whether a point lies inside a zone. A four-layer post-processing stack (exponential moving average smoothing, a dwell timer, an alert cooldown, and the zone modifier) suppresses short-lived pose noise. Its parameters were tuned by hand, and no quantitative ablation of the stack was carried out. For each prediction, the dashboard lists the biomechanical features whose z-scores exceed 1.5 standard deviations, which keeps explanations fast enough for real-time use and avoids the extra computation that SHAP [23] would add.

C. SCRIBE: Voice-to-ADL Documentation

SCRIBE is a modular pipeline that turns caregiver speech into structured Activities of Daily Living (ADL) records and safety alerts. Speech is transcribed, cleaned of disfluencies, and corrected for grammar, then passed to an extraction model that fills in the record and decides whether an alert is needed. Templates produce the written report and a spoken handover.

Speech recognition: We used the Svarah corpus of Indian-accented English [15], which contains 6,656 utterances from 117 speakers. Audio was resampled to 16 kHz, and the utterances were split at random into training, validation, and test sets (80:10:10, seed 42). Whisper-small [14] was adapted with Low-Rank Adaptation (LoRA) [17] in two experiments. The first fixed the LoRA rank at 8 and trained on 10%, 25%, 50%, and 100% of the training set. The second used the full training set and compared ranks 4, 8, 16, and 32. Word error rate (WER) and character error rate (CER) were measured on the same held-out test set for the zero-shot baseline and every adapted model.

Text clean-up: For disfluency removal, DISCO [24] and DisfluencySpeech [25] were combined with 25 domain-specific sentence pairs written for this work, each repeated five times during training. Fine-tuned BART-base and T5-small models were compared using BLEU, ROUGE-L, and a qualitative check on unseen examples. Grammar is then corrected with LanguageTool, with a protected vocabulary of medication and food terms taken from the care-facility information, so that these words are never altered.

Extraction and alerting: The ADL dataset is based on real care information gathered during a facility visit, from discussions with caregivers and their handwritten records. Resident names were replaced with the identifiers P01 to P41 before any processing, and permission to use this information and university ethics approval were obtained. From the de-identified material, natural-language ADL descriptions were generated and expanded synthetically, giving 2,325 unique records in a 19-field schema. T5-small and BART-base were fine-tuned to extract the fields and to predict whether an alert is required, and were compared on alert recall and alert precision. For a stricter check of alerting, a separate set of 200 unseen cases was built and kept out of training: 110 symptom alerts, 20 meal-related alerts, 20 abnormal vital-sign alerts, and 50 non-alert controls. The expected labels were fixed before the model was run and were reviewed by a caregiver.

Output: Reports and handover summaries are generated from fixed, category-specific templates, with no further generative model, and the handover is read aloud using Google Text-to-Speech (gTTS).

D. GLOSS: Personalized Sign Language Tutoring System for Caregivers

GLOSS judges how well a caregiver performs a sign, allows for moderate differences in speed, shows which part of the body deviated, and uses the result to choose further practice.

Data: The WLASL [19] and ASL Citizen [20] corpora were filtered to 60 care-related classes, and one class with too few samples was dropped. Some videos contained the same sign performed several times; these appear as several peaks in motion energy over time rather than one, and were trimmed to a single repetition or removed when the boundary was unclear. This left 2,364 samples across 59 classes, with 22 to 65 samples per class. We divided the samples 70:15:15 for training, validation, and testing, keeping class proportions

the same in each part and splitting at the sample level; signer identity was not controlled, so the same signer may appear in more than one split. MediaPipe Holistic, which builds on BlazePose [13], tracks 49 landmarks per frame. Body-anchored normalization (mid-shoulder translation, shoulder-width scaling, and whole-skeleton rotation) keeps the hands' position relative to the torso. Every sequence is interpolated to a fixed length of 60 frames, whatever its original frame rate, and encoded as landmark positions (147 values per frame) together with velocity and acceleration (441 values in total). Rotation, scaling, and jitter augmentation are applied to the training set only.

Recognition: Three models were compared: a bidirectional long short-term memory network with attention (BiLSTM, about 50K parameters), a SPOTER-style transformer [21] (about 99K), and a temporal convolutional network (TCN) with dilated causal convolutions [22] (about 103K). The BiLSTM was dropped after showing a large gap between training and validation accuracy and low accuracy overall. The SPOTER-style model scored higher than the TCN in a direct comparison (Section IV), but the TCN was kept as the main model because its causal structure suits streaming input and real-time feedback during practice.

Execution assessment: Dynamic Time Warping (DTW) distance alone does not separate signs reliably, so assessment runs in two passes. The first isolates a single attempt and classifies it, and the second re-segments the sequence for the predicted class and scores it. Multivariate DTW (symmetric2 step pattern) compares the attempt with a medoid reference for that class, and each landmark's deviation is expressed as a z-score against that landmark's baseline variation. The overall distance is converted to a quality tier using class-specific thresholds: for each of the 59 classes, the 33rd and 66th percentiles of five raw intra-class DTW distances mark the boundaries between strong, moderate, and weak.

Feedback: A rule-based generator turns the largest deviations into body-part-specific feedback. A large language model (LLM), Gemini (gemini-3.5-flash-lite), then rephrases it in plain language, using the predicted sign, quality tier, and landmark deviations as context; it does not decide the tier or the deviations. A fixed prompt restricts it to rephrasing: it may not add claims, body parts, or advice, must keep to two or three short sentences, and may use praise words only for the strong tier (temperature 0.6). An automated check after every generation flags landmark names absent from the input and praise words in weak or moderate tiers. Lessons are then selected according to the errors identified.

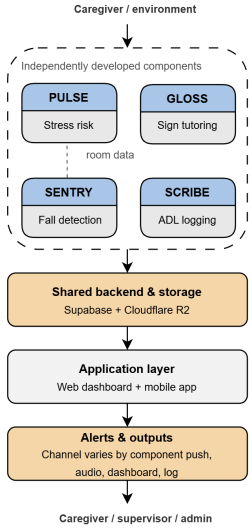


Fig. 1. System architecture of the proposed platform, showing the four independently developed components, shared backend, application layer, and alert outputs.

IV. Results and Discussion

A. PULSE

Table I reports the final model. On the 27 test caregivers (1,406 prompts), it reached a ROC-AUC of 0.7211 (95% CI 0.6644 to 0.7699), and repeated grouped cross-validation gave a mean of 0.6991 (SD 0.0294, range 0.6344 to 0.7655). Test accuracy (0.7930) was close to the 0.7873 obtained by always predicting non-elevated stress, so accuracy is not informative here. Causal features alone gave 0.5361. Adding lag-1 stress raised this to 0.6526 but caused overfitting (training ROC-AUC 0.8654), regularization and early stopping narrowed the gap (0.7086), and the learned heart-rate baseline gave the final 0.7211. With steps deviation rebuilt from earlier days only, test ROC-AUC was 0.7213.

Lag-1 stress dominated the SHAP ranking (mean absolute value 0.208), ahead of maximum heart rate (0.068) and the 7-day heart-rate deviation (0.047), and six features received no weight. On the 1,379 test prompts with a previous response, lag-1 stress alone reached 0.7026 against 0.7273 for the full model, a difference of +0.0247 (95% CI -0.0010 to $+0.0515$). In the proximity ablation (Table II), all six models scored between 0.4345 and 0.4604, and every CI of the change from adding centrality included zero. Stress did not differ between high- and low-centrality days ($t = -1.52$, $p = 0.129$, $d = -0.061$). In the daily records, elevated stress showed a weak association with personal heart-rate deviation ($r = 0.024$, $p = 0.017$), none with raw heart rate ($r = 0.005$, $p = 0.625$), and a stronger one with lower sleep efficiency ($r = -0.106$, $p < 0.0001$).

Stress risk is therefore predictable above chance from data available before the report, but most of the signal comes from the caregiver's own previous response. The gain from the wearable features is small and its CI just reaches zero, so a larger test cohort is needed to confirm it; the small final

model (11 trees) is consistent with a weak daily signal. The 0.84 reported on TILES-2019 [5] came from different participants and labels, so the figures are not directly comparable. Centrality showed no detectable benefit rather than harm, since neither feature set beat chance on the small matched subset. The heart-rate association is very weak, and pooling repeated records from the same caregivers can overstate its significance. All results come from a single cohort, so generalization to other caregivers, devices, or care settings remains untested.

TABLE I
FINAL PULSE MODEL PERFORMANCE

Evaluation	ROC-AUC	Accuracy
Train	0.7328	0.7691
Validation	0.6874	0.6966
Test (95% CI)	**0.7211** (0.6644–0.7699)	0.7930 (0.7265–0.8591)
Repeated grouped CV (25 folds, mean ± SD)	0.6991 ± 0.0294	0.7489 ± 0.0387

TABLE II
ABLATION: ADDING PROXIMITY CENTRALITY (MATCHED SUBSET, TEST ROC-AUC)

Model	Set A	Set B	Δ (B−A) [95% CI]
Random Forest	0.4576	0.4548	−0.0028 [−0.0379, +0.0361]
XGBoost	0.4604	0.4460	−0.0144 [−0.0506, +0.0269]
LightGBM	0.4434	0.4345	−0.0089 [−0.0487, +0.0307]

B. SENTRY

Under the subject-disjoint protocol, the late-fusion model reached 97.12% accuracy, 92.31% sensitivity (72 of 78 falls detected across 312 test windows), a 1.28% false-alarm rate, and a ROC-AUC of 0.9836 (Table III). The random clip-level split gave 98.27% accuracy and a 0.70% false-alarm rate. The accuracy gap between the two protocols is small (1.15 percentage points), but the false-alarm rate nearly doubles when test subjects are unseen, which is consistent with some inflation from identity leakage in the random split.

Both branches contribute. On the same subject-disjoint split, the Random Forest on the 18 biomechanical features reached 94.56% accuracy (macro-F1 0.9349) and the ST-GCN alone 96.02% (macro-F1 0.9492), against 97.12% for late fusion. Adding the biomechanical branch to the ST-GCN also lowered the false-alarm rate from 4.41% to 1.28%.

Transfer across datasets was poor. Trained on NTU RGB+D and tested on UR Fall Detection, the model's ROC-AUC fell to 0.410. A value below 0.5 means normal clips were ranked above falls more often than not, which suggests a systematic shift between the Kinect v2 and MediaPipe skeleton pipelines rather than random error. Fine-tuning the last two fusion layers on 35 webcam clips from one subject partly recovered performance on 58 clips from a second subject, with a macro-F1 of 0.8851, accuracy of 91.38%, sensitivity of 80.00%, and false-alarm rate of 4.65%.

On an AMD Ryzen 7 8845HS CPU, one 90-frame window took 4.515 ms, or 221.5 windows per second. Real-time use needs 10 inferences per second, one every three frames at 30 fps.

These results support using SENTRY with RGB webcams placed much like those in the training and adaptation data, but not across different cameras, rooms, or patient groups. The webcam test involved a single subject and one camera setup, so it shows adaptation to that setting rather than general robustness. Its 80.00% sensitivity also means one in five falls in that setting was missed, which matters for a safety tool.

TABLE III
SENTRY FALL-DETECTION EVALUATION RESULTS

Test	Key Result(s)	Meaning
Subject-disjoint (late fusion)	Accuracy 97.12%; sensitivity 92.31%; false alarm 1.28%; ROC-AUC 0.9836	Main protocol (unseen subjects)
Random split	Accuracy 98.27%; false alarm 0.70%	Upper bound (subject overlap)
Baselines	RF: accuracy 94.56%, macro-F1 0.9349; ST-GCN: accuracy 96.02%, macro-F1 0.9492	Each branch on its own
Cross-dataset (NTU → UR)	ROC-AUC 0.410	Drop on a new dataset
Webcam test (58 clips)	Macro-F1 0.8851; accuracy 91.38%; sensitivity 80.00%; false alarm 4.65%	Partial recovery
Biomech. branch	False alarm reduced: 4.41% → 1.28%	Fewer false alarms
CPU speed	4.515 ms/window; 221.5 windows/s	Meets real-time need

C. SCRIBE

Speech recognition: Table IV summarizes the recognition experiments. Zero-shot Whisper-small reached a WER of 26.13% and a CER of 12.85% on the 666 test utterances. With the LoRA rank fixed at 8, WER fell steadily as more training data was added, from 19.51% with 10% of the data to 15.57% with all of it. Across ranks, rank 32 gave the lowest WER (14.86%), but rank 16 came close (15.35%) with about half the trainable parameters, so it was kept. Retrained as a separate run for the final model, rank 16 reached a WER of 15.57% and a CER of 5.32%, a 40.4% relative reduction in WER from the zero-shot baseline. For context, a Whisper fine-tuning study on Indian-accented English reported 17.36% WER [16]; it used different data and recording conditions, so the two values are not directly comparable and no claim of improvement is made.

Disfluency removal: On the general test set, T5-small scored slightly higher than BART-base (BLEU 0.9616 against 0.9530; ROUGE-L 0.9856 against 0.9817). BART-base was chosen nevertheless, because in the qualitative check it handled a safety-relevant self-correction, in which the speaker first gave a wrong dose and then corrected it, better than T5-small.

Extraction and alerting: On the original held-out split, T5-small detected all 54 alert-positive cases (alert recall 100%), against 70.37% for the best BART-base run. On the separate 200-case safety set (Table V), its alert recall fell to 82.00%, with 96.09% precision, an F1-score of 88.49%, and 84.00% accuracy. Of the 27 missed alerts, 23 were symptom-related, 3 meal-related, and 1 vital-sign-related, which gives a recall of

79.1% for symptom alerts, 85.0% for meal alerts, and 95.0% for vital-sign alerts. One missed case described dizziness, which none of the training categories covered.

The fall from 100% to 82.00% shows that a single test split can overstate how reliably alerts are caught, and symptom descriptions, whose wording varies most, were the hardest. Several factors limit these results. The Svarah split was made by utterance, so test speakers may also appear in training, and the WER may be optimistic for new speakers. Most ADL records are synthetic expansions of one facility's de-identified information, so their language may be more regular than real caregiver notes. BART-base was selected for disfluency removal on the basis of a single qualitative example. Larger evaluations with real caregiver observations are needed to understand false negatives and how well the system generalizes.

TABLE IV
SCRIBE ASR BASELINE, ABLATION, AND FINAL MODEL RESULTS

Experiment	Configuration	WER (%)	CER (%)
Baseline	Whisper-small zero-shot	26.13	12.85
Data efficiency ($r = 8$)	10% data	19.51	6.20
	25% data	17.32	5.78
	50% data	16.31	5.49
	100% data	15.57	5.22
Rank ablation (100% data)	$r = 4$	16.04	5.34
	$r = 8$	15.70	5.19
	$r = 16$ (selected)	**15.35**	**5.10**
	$r = 32$	14.86	5.15
Final model	**$r = 16$ retrained**	**15.57**	**5.32**

TABLE V
SCRIBE SAFETY-FOCUSED ALERT EVALUATION

Metric	Result
Total cases	200
True positives / false negatives	123 / 27
False positives / true negatives	5 / 45
Alert recall / precision	82.00% / 96.09%
F1-score / accuracy	88.49% / 84.00%
Recall: symptom alerts	87/110 (79.1%)
Recall: meal-related alerts	17/20 (85.0%)
Recall: vital-sign alerts	19/20 (95.0%)

D. GLOSS

Recognition: The TCN reached a test macro-F1 of 0.75 (accuracy 0.76, macro-precision 0.77, macro-recall 0.75; $n = 355$). This is the run closest to the mean of four independent training runs (mean 0.735, SD 0.044), reported so that the figure is typical rather than the best case. Under the same augmented kinematic configuration, the SPOTER-style transformer scored higher than the TCN in a single-run comparison, on both validation data (macro-F1 0.80 against 0.70) and test data (0.74 against 0.66). The TCN's four-run mean narrows this gap but does not show an accuracy advantage. Accuracy was uneven across the vocabulary: signs with few samples and little movement were recognized inconsistently between runs, possibly because their velocity and acceleration channels carry little signal.

Execution assessment: On the raw DTW scale used by the deployed thresholds, the mean intra-class distance was 822.20 (SD 659.52, $n = 295$) and the mean inter-class distance 1079.56 (SD 578.96, $n = 59$). Pooled across classes, 69.49% of comparisons against the wrong class's reference still fell below the intra-class mean plus one standard deviation, that is, within the range expected for a correct sign. DTW distance on its own therefore cannot tell a correct sign from a different one, which is why GLOSS recognizes the sign first and scores execution only against that sign's reference. In the speed test, 15 real recordings from randomly chosen classes were stretched or compressed to between $0.7\times$ and $1.3\times$ of their original duration. DTW distance to the original rose smoothly and monotonically with the change in speed, with a normalized swing of 0.337 across the range, so moderate tempo differences are penalized gradually rather than sharply.

Feedback faithfulness: The automated check passed 91.5% of 59 LLM-rephrased outputs across the three quality tiers (20 of 21 strong, 21 of 22 moderate, and 13 of 16 weak). When the team inspected the five flagged outputs by hand, all turned out to be false alarms of the keyword matching: four came from a low-confidence fallback message that already mentions the hands, and one from the sign "head" being read as a body part. No praise words appeared in weak or moderate tiers.

Recognition is moderate for a 59-class vocabulary, and the SPOTER-style model remains a strong alternative if streaming is not required. Because the split was made at sample level without controlling for signer identity, the same signer can appear in both training and test data, so these figures may be optimistic for new signers. The DTW calibration samples and the 59 faithfulness attempts were drawn from the full preprocessed corpus rather than from the test split alone, and the per-class thresholds rest on only five distances each. The faithfulness check and its manual review were carried out by the development team, so they are not an independent validation, and the effect of the feedback on learning has not been measured with learners. Both remain future work.

TABLE VI
GLOSS RECOGNITION AND EXECUTION-EVALUATION RESULTS SUMMARY

Metric	Value
TCN test macro-F1 (reported run, $n = 355$)	0.75
TCN macro-F1 over 4 runs (mean $\pm$ SD)	0.735 ± 0.044
SPOTER-style test macro-F1 (single run)	0.74
DTW intra-class distance, raw (mean $\pm$ SD, n)	822.20 ± 659.52 (295)
DTW inter-class distance, raw (mean $\pm$ SD, n)	1079.56 ± 578.96 (59)
Inter-class distances within intra-class mean + 1 SD	69.49%
Speed-tolerance distance swing ($0.7\times$–$1.3\times$)	0.337
LLM-phrasing faithfulness pass rate ($n = 59$)	91.5%

V. CONCLUSION

This paper asked how well four machine learning modules, built for one caregiver platform, perform on people or data they did not see during training. PULSE predicted stress

risk from pre-survey data with a test ROC-AUC of 0.7211 (95% CI 0.6644 to 0.7699) and 0.6991 in repeated grouped cross-validation. Most of its signal came from the caregiver's previous stress report, and sensor-measured proximity added no detectable value. SENTRY reached 97.12% accuracy at a 1.28% false-alarm rate on unseen subjects, but its ROC-AUC fell to 0.410 on a different dataset; fine-tuning on one subject's webcam clips recovered a macro-F1 of 0.8851 on another. SCRIBE achieved a WER of 15.57%, and its alert recall dropped from 100% on the original split to 82.00% on a separate safety set, mostly through missed symptom alerts. GLOSS recognized signs with a macro-F1 of 0.75 and located execution errors at landmark level. Where both easier and stricter tests were run, the stricter ones gave lower but more realistic figures, and they show where each module is still weak. No end-to-end evaluation of the combined platform was carried out. Next steps are to test PULSE on other caregiver cohorts and to evaluate SENTRY across more cameras and patients. SCRIBE needs broader symptom coverage and a larger safety evaluation, and the GLOSS feedback, including its LLM-rephrased output, needs independent review and a study with learners.

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